

IoT Based Smart Light Automation using ESP32 and Blynk

Objective

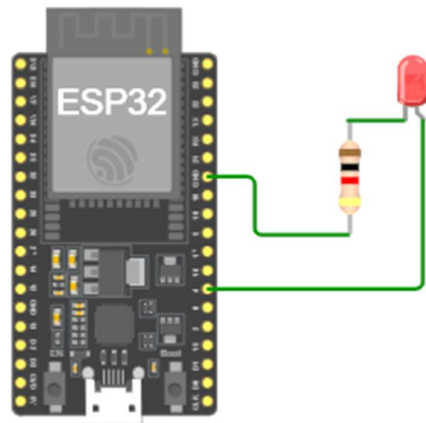
The objective of this project is to develop an IoT-based smart light automation system using ESP32 and Blynk cloud platform. The system enables remote wireless control of an LED through a mobile application using WiFi communication.

Software Used

Software	Purpose
Arduino IDE	Programming ESP32
Blynk IoT Platform	Cloud communication
Wokwi Simulator	Virtual hardware simulation

Circuit Connections

LED Pin	ESP32 Pin
LED Positive (Anode)	GPIO 4
LED Negative (Cathode)	GND through 220Ω resistor



Working Principle

The ESP32 connects to the internet using WiFi and communicates with the Blynk cloud platform. A button widget is created in the Blynk mobile application. When the user presses the button, a command is sent through the cloud to the ESP32. The ESP32 receives the command and controls the LED connected to GPIO pin 4. The LED turns ON or OFF based on the received command, demonstrating real-time IoT automation.

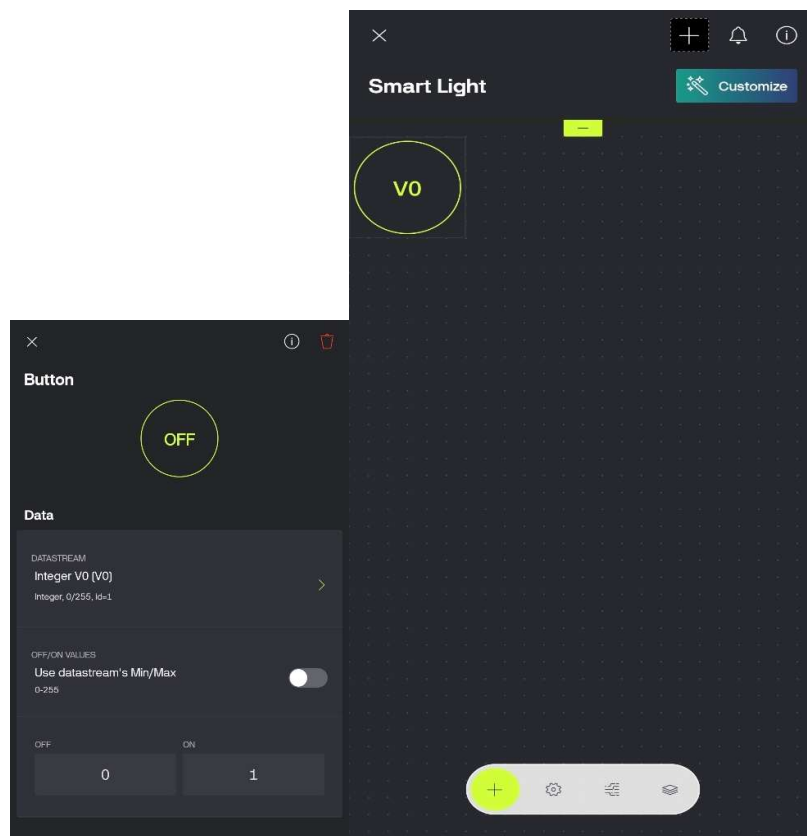
Source Code

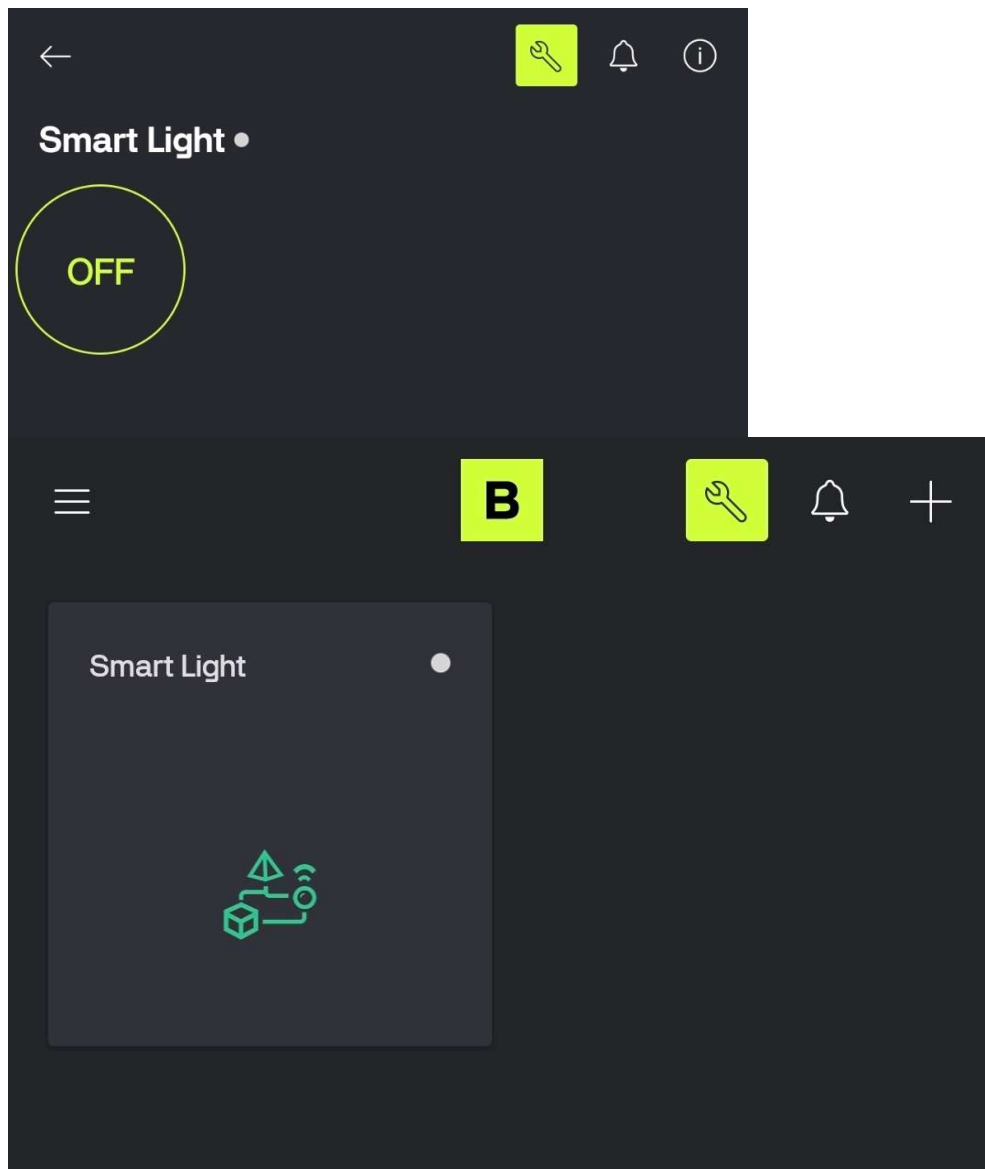
```
1  #define BLYNK_TEMPLATE_ID "TMPL30mtsdoEo"
2  #define BLYNK_TEMPLATE_NAME "Smart Light"
3  #define BLYNK_AUTH_TOKEN "_SEwsb776pRZ77qQibqqfbPGchX4IyTG"
4
5  #include <WiFi.h>
6  #include <WiFiClient.h>
7  #include <BlynkSimpleEsp32.h>
8
9  char ssid[] = "Wokwi-GUEST";
10 char pass[] = "";
11
12 #define LED_PIN 4
13
14 BLYNK_WRITE(V0)
15 {
16     int value = param.asInt();
17
18     if (value == 1)
19     {
20         digitalWrite(LED_PIN, HIGH);
21         Serial.println("LED ON");
22     }
23     else
24     {
25         digitalWrite(LED_PIN, LOW);
26         Serial.println("LED OFF");
27     }
28 }
29
30 void setup()
31 {
32     Serial.begin(115200);
33
34     pinMode(LED_PIN, OUTPUT);
35
36     Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
37
38     Serial.println("Connected to Blynk");
39 }
40
41 void loop()
42 {
43     Blynk.run();
44 }
```

Code Explanation

- Blynk credentials (Template ID, Template Name, and Auth Token) are used to connect ESP32 with the Blynk cloud platform.
- WiFi library enables ESP32 to connect to the internet using WiFi.
- LED is connected to GPIO pin 4 of ESP32.
- BLYNK_WRITE(V0) function receives data from the Blynk mobile app button widget.
- When button value becomes 1, ESP32 turns the LED ON.
- When button value becomes 0, ESP32 turns the LED OFF.
- In setup(), serial communication starts, LED pin is configured as output, and ESP32 connects to Blynk cloud.
- In loop(), Blynk.run() continuously maintains communication between ESP32 and Blynk server.
- The project enables real-time wireless LED control using IoT technology.

BLYNK SETUP





Output

- ESP32 successfully connected to Blynk cloud platform.
- LED controlled remotely through Blynk mobile application.
- Real-time wireless ON/OFF operation achieved using IoT.

```
Connected to Blynk
LED ON
LED OFF
LED ON
LED OFF
LED ON
LED OFF
```